

Product Requirements Document (PRD)

Project Name: Condominium Management Authority (CMA) Management System

Date: Jun 16 ,2026

Method : React/Node js express/Sql lite

1.0 Executive Summary

The CMA Management System is a comprehensive, web-based platform designed to digitize, centralize, and streamline the core operational workflows of the Condominium Management Authority (CMA) of Sri Lanka. The system aims to replace disparate spreadsheet-based records with a unified, secure, and efficient application that improves data management, enhances visibility, and facilitates regulatory compliance.

This document outlines the functional requirements, user interface design, and technical specifications for the development of the CMA Management System. The system will empower CMA staff to manage Management Corporations (MCs), Management Committees (M.Coms), C-Files, and complaints with enhanced efficiency, transparency, and reporting capabilities.

2.0 Project Goals & Objectives

1. **Digital Transformation:** Transition from manual, spreadsheet-driven processes to a centralized, secure, and intuitive digital system.
2. **Data Centralization:** Create a single source of truth for all MC, [M.Com](#), C-File, and complaint records, Discussion, Registration Form -MC eliminating data silos and ensuring data integrity.
3. **Improved Workflow Efficiency:** Automate and streamline key workflows, particularly the complaint resolution process, including escalation and approval stages.
4. **Enhanced Visibility & Tracking:** Provide a comprehensive dashboard and powerful reporting tools to give management real-time insights into operational status, pending tasks, and overall performance metrics.
5. **Regulatory Compliance:** Ensure all system processes align with the requirements of the Condominium Management Authority.

3.0 User Roles & Personas

The system will be built around the following core user roles, with potential for future expansion:

1. **Super Administrator:** Full system access. Manages user accounts, system configuration, and has oversight of all modules. Responsible for the overall integrity of the system.
2. **DGM/Administrator (Deputy General Manager/Administrative Head):** Oversees departmental operations. Has full access to all records, can approve and escalate cases, generate reports, and manage users within their team.
3. **CCU Officer (Complaints/Examining Officer):** Handles day-to-day tasks. Can add, edit, view, and update records. Initiates the complaint workflow and conducts the initial officer check.
4. **DGM (Approver):** A specific role for senior staff who are responsible for the second stage approval in the complaint workflow.
5. **Legal Officer:** A specific role responsible for the final review and closure of complaints that have been escalated to the legal department.

4.0 Functional Requirements

4.1 General Application Features

- **User Authentication & Authorization:** Secure login with role-based access control to restrict access to sensitive data and functions.
- **Navigation:** A clear and intuitive left-side navigation menu to access all major modules.
- **Global Search:** A persistent top-bar search function to quickly find any record across the entire system.
- **Notifications:** An alert system for logged-in users to be notified of pending tasks, especially new or overdue complaints.
- **Audit Logging (Last Modification):** Automatic tracking of changes made to any record. Each row must capture the **user who made the modification** and the **timestamp (date & time)** of the change. This is a critical requirement for compliance and accountability. This feature will be applied at the row-level for all data tables (MC, [M.Com](#), C-Files, Complaints).

- **Data Management:**

- **Create:** Users with appropriate permissions can add new records to the system.
- **Read (View):** Users can view detailed records.
- **Update (Edit):** Users with appropriate permissions can modify existing records. "Create by" and "Last Modification" audit log will be updated accordingly.
- **Delete:** Users with appropriate permissions can delete records (likely with a soft-delete or confirmation prompt).
- **Export:** All module pages will allow users to **export the current filtered dataset to an Excel file** (.xlsx).
- **Import:** All module pages will allow users to **import data from an Excel file** to bulk-add new records or update existing ones.

4.2 Module 1: Dashboard

- **Objective:** Provide a high-level, at-a-glance view of the system's status.
- **Requirements:**
 - **Key Performance Indicators (KPIs):**
 - Total count of MCs, M.Coms, C-Files, and Complaints, Registration Form, Discussion, Records, Report Generate
 - Breakdown by status (Active, Non-Active, Open, Pending, Closed, etc.).
 - Highlight complaints pending for over 30 days with a visual alert (e.g., red color or fire icon).- 15 day normal reminder for the dashboard 30day after highlight the pending
 - **Visualizations:**
 - Charts showing monthly trends for records and complaints.
 - Status distribution of complaints (Open, Pending, Closed, Resolved) in a pie/bar chart.
 - **Recent Activity:** A feed of the latest system activities (e.g., record updates, new complaints filed, complaint approvals).

4.3 Module 2: MC Management

- **Objective:** Manage all records related to incorporated Management Corporations (MCs).

- **Data Source:** 05 - MC List - May.xlsx
- **Requirements:**
 - **Display:** A tabular view of all MC records with all fields from the source Excel sheet.
 - **Actions:** Create, View, Edit, Delete, Import, Export.
 - **Filters:**
 - **Category/Year:** Search by year of registration.
 - **General Search:** Search across all fields.
 - **Status:** Filter by status (Active, Non-Active, Pending).
 - **Field Mapping:** The database table for MCs will include all columns listed in the 05 - MC List - May.xlsx file.
 - **Key Fields:**
 - Reg_Date, Old_File_No, New_File_No, Management_Corporation_Name, Address, Plan_No, Units, Residential, Non_Residential, Service_Units, Category, Renewal_Period, AGM_Date, Next_AGM_Date, Renewal_Date_Vise, AGM_Date_Vise, MC, Engineer, MA, Secretary, Secretary_Unit_No, Treasurer, Treasurer_Unit_No, AGM_Minutes, Attendance, Audited_Accounts, Final_Accounts, Building_Insurance, Sinking_Fund, Budget_Proposal, Town, Municipal_Council_Pradeshia_Saba, Certificate_Division_File_No, Land_Registry_Approved_Date, Declaration_No, Email_Address.

4.4 Module 3: [M.Com](#) Management

- **Objective:** Manage all records related to non-incorporated Management Committees (M.Coms).
- **Data Source:** M.Com List.xlsx
- **Requirements:**
 - **Display:** A tabular view of all [M.Com](#) records.
 - **Actions:** Create, View, Edit, Delete, Import, Export.
 - **Filters:**
 - **Category/Year:** Search by year of registration.

- **General Search:** Search across all fields.
- **Status:** Filter by status (Active, Non-Active, Pending).
- **Field Mapping:** The database table for M.Coms will include all columns listed in the M.Com List.xlsx file.
- **Key Fields:**
 - reg_date, Old_File_No, New_File_No, Management_Corporation_Name, Address, Units, Plan_No, Secretary, unit_No, Treasurer, unit_No, Renewal_Period, AGM_Date, AGM_Minutes, Attendance, Audited_Account, Building_Insurances, Renewal_Date_Vise, AGM_Date_Vise, MC / M.Com, Eng., MA, Town, Municipal_Council/Pradeshiya_Saba_limits, Certificate_Division_File_No., Email_Address, Awareness_Date.

4.5 Module 4: C Files

- **Objective:** Manage the registry of official C-Files.
- **Data Source:** c file & complaint page catog.jpg and C file - register 19.03.2024..xlsx
- **Requirements:**
 - **Display:** A tabular view of all C-File records.
 - **Actions:** Create, View, Edit, Delete, Import, Export.
 - **Filters:**
 - **General Search:** Search across all fields.
 - **Status:** Filter by status (Open, In Progress, Closed, Urgent).
 - **Field Mapping:** The database table for C Files will include the fields from the C file - register 19.03.2024..xlsx and the categories from c file & complaint page catog.jpg.
 - **Key Fields:**
 - File_No, Name_of_the_Apartment, Address, Date, Reason, Complainer_Details, Registered_CCU_File_No, Remarks. Also include a field for "**Category**" which should be a select list with options:
 - Non Payment
 - Pet Keeping

- Parking Issues
- Accounts related Issues
- Council Issues
- Garbage Issues
- Common Elements Issues
- Other Issues
- Unauthorised Construction
- Water Leakage
- Sinking Fund
- Common area issue
- Boundary Issues
- Developer Related Issues (Construction)
- Developer Related Issues (Legal Issue)
- Common Elements Issue (Developer)
- Non registration of Condominium Plan
- Business Purpose

4.6 Module 5: Complaints Management

- **Objective:** Manage the lifecycle of public complaints, from filing to resolution, with a structured, multi-stage approval workflow.
- **Requirements:**
 - **Workflow Definition:** The system MUST implement the following multi-stage workflow for complaint processing:
 1. **Complaint Filed (filed):** The initial state when a new complaint is logged.
 2. **Officer Check (check):** An assigned officer reviews the complaint. This is the first action point.
 3. **DGM Approval (dgm):** After the officer check, the complaint is escalated to the DGM for approval and decision.

4. **Legal Review (legal):** After DGM approval, the complaint is sent to the Legal Section for final review and closure (legal_closed).
 - **Display:** A tabular view of all complaints, with visual indicators for status (e.g., colors for Open/Pending/Closed).
 - **Actions:** Create, View, Edit, Delete, Import, Export.
 - **Filters:**
 - **General Search:** Search across all fields.
 - **Status:** Filter by status (Open, Pending, Resolved, Closed).
 - **Category:** Filter by category (Technical, Legal, Others).
 - **Field Mapping:** The database table for Complaints will include all necessary fields to support the workflow.
 - **Key Fields:**
 - Ref_No, Complainant, Building, Subject, Filed, Category, Status, Assigned_To, DGM_Match, Stage, Pending_Reason, Urgent_Reason, Comments, Closed_By, Approved_By.
 - **Workflow**
Fields: wfStage, wfCheckBy, wfCheckDate, wfDGMBY, wfDGMDDate, wfDGMNotes, wfLegalBy, wfLegalDate, wfLegalNotes, wfLegalClosed.
 - **UI Considerations:**
 - **Pending Complaints Alert:** On the Dashboard and in the Complaint List, clearly highlight complaints pending for more than 30 days.
 - **Pending Detail Panel:** A dedicated, expandable section to view a list of all pending complaints, showing key details, pending reasons, and urgency flags.
 - **Action Buttons:** Contextual buttons for each stage of the workflow (e.g., "Officer Check", "DGM Approve", "Legal Close").

4.7 Module 6: Registration Form

- **Objective:** Digitize the "Application for Registration as a Management Corporation" form.
- **Data Source:** Registration form.jpg and Registration form 2.jpg

- **Requirements:**
 - **Form Fields:** Replicate the paper form digitally with all required fields:
 - **Section 1:** Title Name, Registered Condominium Plan No, Official Address.
 - **Section 4:** Number of parcels (Residential/Commercial), Category.
 - **Section 5:** Written Assurance (checkbox/statement), list of Annexes (checkboxes).
 - **Section 6:** Services & Facilities (checkboxes like Lift, Gym, etc.).
 - **M/C Controlled by Management Company:** Radio buttons and conditional fields (Company Name, Contact No).
 - **Action:** "Save Application" button to submit the form and create a pending application record.

4.8 Module 7: Records

- **Objective:** A unified view of all records across all modules.
- **Requirements:**
 - **Tab View:** Tabs for "All Records", "MC", "[M.Com](#)", "C Files", and "Complaints".
 - **Display:** A combined, filterable table showing key information from all modules.
 - **Actions:** Quick action buttons to Edit or Delete any record.

4.9 Module 8: Report Generator

- **Objective:** Allow users to generate and download/print reports.
- **Requirements:**
 - **Pre-defined Reports:** One-click report generation for each module (MC, [M.Com](#), C-Files, Complaints).
 - **Full Summary Report:** A comprehensive report covering all modules.
 - **Preview:** Show a preview of the report before downloading or printing.
 - **Export Options:** "Export PDF" and "Print" functionalities.

4.10 Module 9: My Profile

- **Objective:** Allow users to view and manage their profile.

- **Requirements:**
 - **View Profile:** Display the user's personal and contact details.
 - **Change Password:** A form to change the user's password.
 - **Edit Profile:** Option to edit profile details (Name, Email, Phone).
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5.0 Technical Specifications

- **Architecture:** Single-page application (SPA) using HTML, CSS, and JavaScript.
 - **Frontend Framework:** Vanilla JS to ensure accessibility and minimal dependencies.
 - **Styling:** Custom CSS3 with a clean, professional UI.
 - **Data Storage:** localStorage for client-side data persistence.
 - **Data Handling:**
 - **Import/Export:** Use the XLSX library to handle Excel file operations.
 - **PDF Export:** Use the jspdf library and its jspdf-autotable plugin for PDF generation.
 - **Version Control:** N/A for this prototype, but would require Git for real project development.
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6.0 Non-Functional Requirements

1. **Performance:** The application should load quickly and be responsive to user interactions. The use of localStorage will ensure near-instantaneous data access for the prototype.
2. **Usability:** The interface should be intuitive and easy to navigate for users of varying technical skills. Clear labels, consistent layouts, and visual feedback are essential.
3. **Reliability:** The system should handle errors gracefully and prevent data loss. The "Last Modification" audit log will help track data changes.
4. **Security:**
 - **Authentication:** A secure login mechanism is required.
 - **Authorization:** Role-based access control (RBAC) must be implemented to ensure users only have access to the data and functions appropriate for their role.

- **Data Validation:** Input validation and sanitization to prevent injection attacks.
 - 5. **Maintainability:** Clean, well-commented code and a modular design to facilitate future enhancements and bug fixes.
 - 6. **Auditability:** The system must provide a detailed audit trail at the record level, capturing which user made a change and when. This is a key requirement for compliance and transparency.
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7.0 User Interface (UI) Guidelines

- **Color Palette:** Use the CMA's official blue (#1a3a6b and #2251a3) and gold (#c9a84c and #e8c96a) as primary brand colors. Use neutral colors for backgrounds and secondary elements.
- **Typography:** Use a modern, readable font like DM Sans for body text and Playfair Display for headings or special elements.
- **Layout:**
 - **Fixed Sidebar:** The navigation menu should be on the left side.
 - **Topbar:** Contains the page title, global search, and user profile/notification area.
 - **Main Content Area:** The primary area for displaying data and forms.
- **Visual Feedback:** Use subtle animations, color changes, and toast notifications to provide clear feedback to user actions.
- **Responsiveness:** Ensure the UI is functional and visually acceptable on different screen sizes, prioritizing desktop and tablet use for an administrative tool.